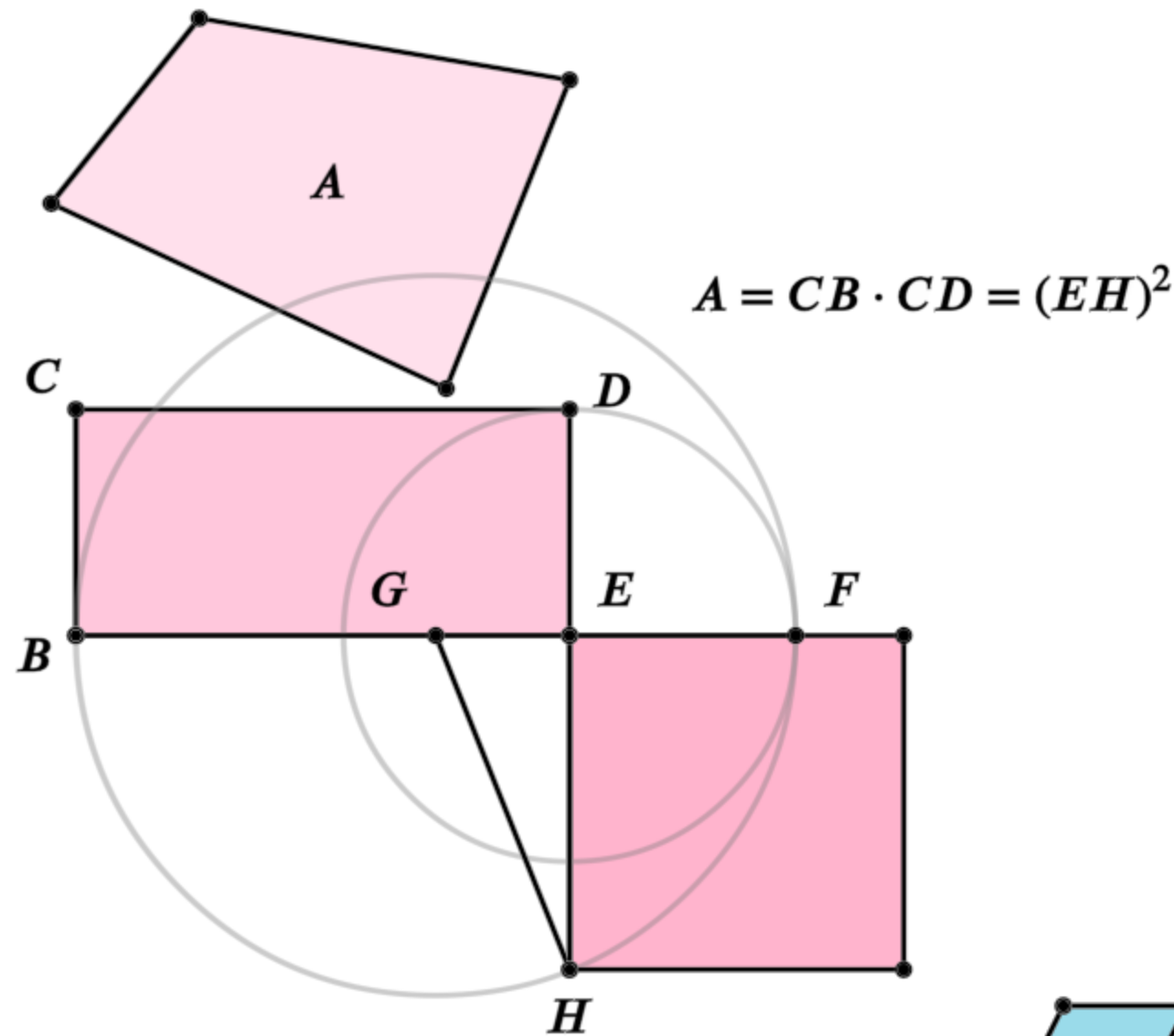


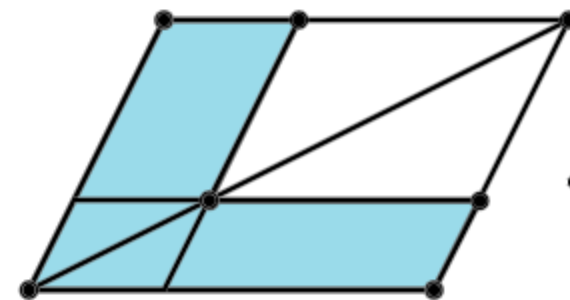
Euclid's Elements

Book II



It is a remarkable fact in the history of geometry, that the Elements of Euclid, written two thousand years ago, are still regarded by many as the best introduction to the mathematical sciences.

- Florian Cajori,
A History of Mathematics (1893)

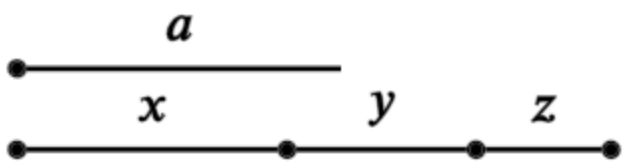


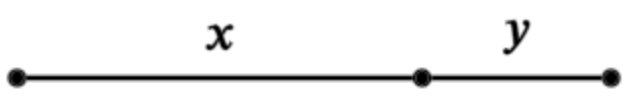
Definitions:

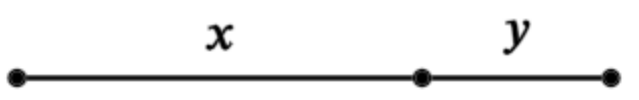
Any rectangular parallelogram is said to be contained by the two straight lines containing the right angle.

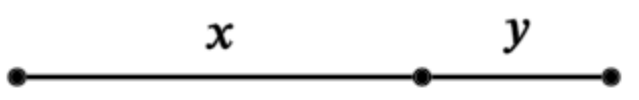
And in any parallelogrammic area let any one whatever of the parallelograms about its diameter with the two complements be called a gnomon.

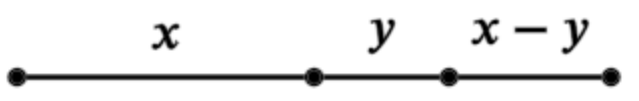
Table of Contents - Book 2

1. 
 $a(x + y + z) = ax + ay + az$

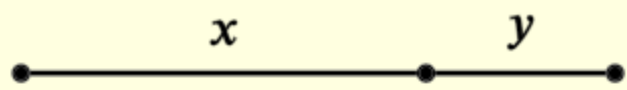
2. 
 $(x + y)^2 = x(x + y) + y(x + y)$

3. 
 $y(x + y) = yx + y^2$

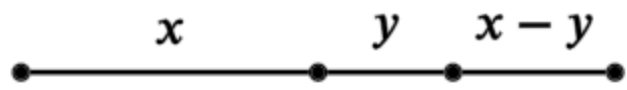
4. 
 $(x + y)^2 = x^2 + y^2 + 2xy$

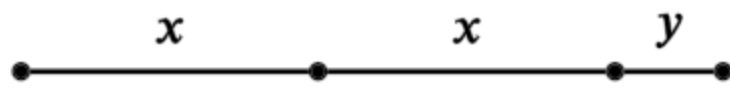
5. 
 $(x + y)(x - y) + y^2 = x^2$

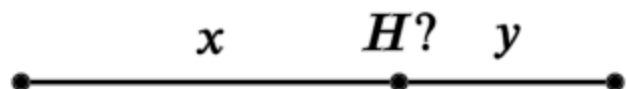
6. 
 $y(2x + y) + x^2 = (x + y)^2$

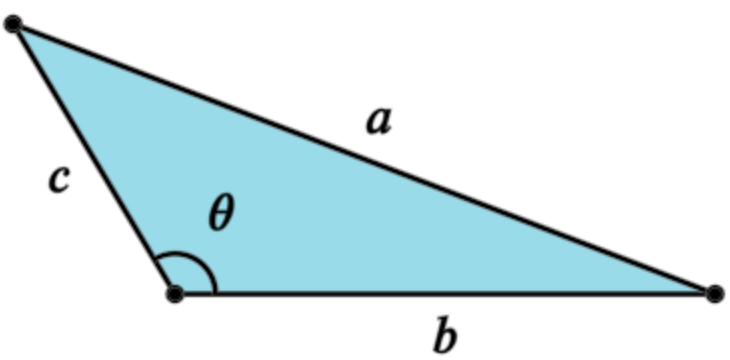
7. 
 $(x + y)^2 + y^2 = x^2 + 2y(x + y)$

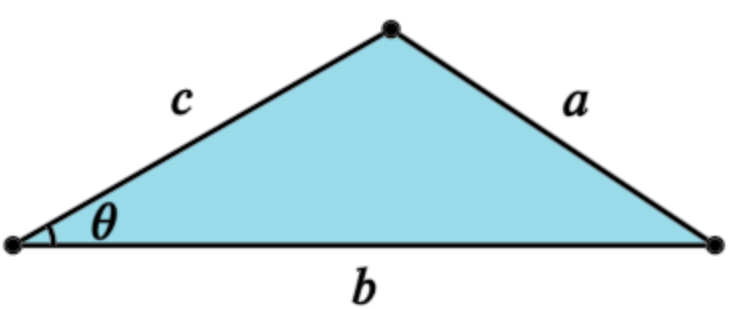
8. 
 $(x + 2y)^2 = 4y(x + y) + x^2$

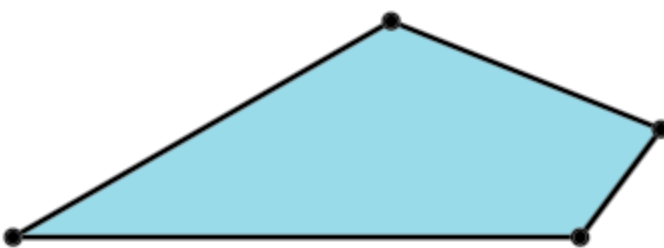
9. 
 $(x + y)^2 + (x - y)^2 = 2(x^2 + y^2)$

10. 
 $(2x + y)^2 + y^2 = 2(x^2 + (x + y)^2)$

11. 
Find H such that: $y(x + y) = x^2$

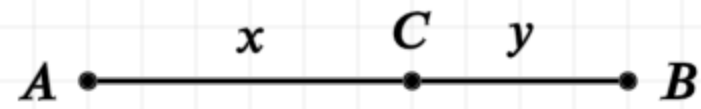
12. 
Cosine Law, $a^2 = b^2 + c^2 - 2bc \cos \theta$

13. 
Cosine Law. $a^2 = b^2 + c^2 - 2bc \cos \theta$

14. 
Construct a square equal to the polygon

Proposition 7 of Book 2

If a straight line be cut at random, the square on the whole and that of one of the segments both together are equal to twice the rectangle contained by the whole and the said segment and the square on the remaining segment.



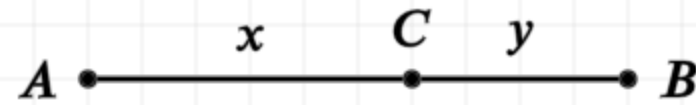
In other words

Let *AB* be a straight line, arbitrarily cut at point *C*

$$AB = AC + CB$$

Proposition 7 of Book 2

If a straight line be cut at random, the square on the whole and that of one of the segments both together are equal to twice the rectangle contained by the whole and the said segment and the square on the remaining segment.



In other words

Let *AB* be a straight line, arbitrarily cut at point *C*

Then the squares formed by lines *AB* and *BC* are equal in area to the sum of the square formed by line *AC*, plus twice the area of the rectangle formed by lines *AB* and *CB*

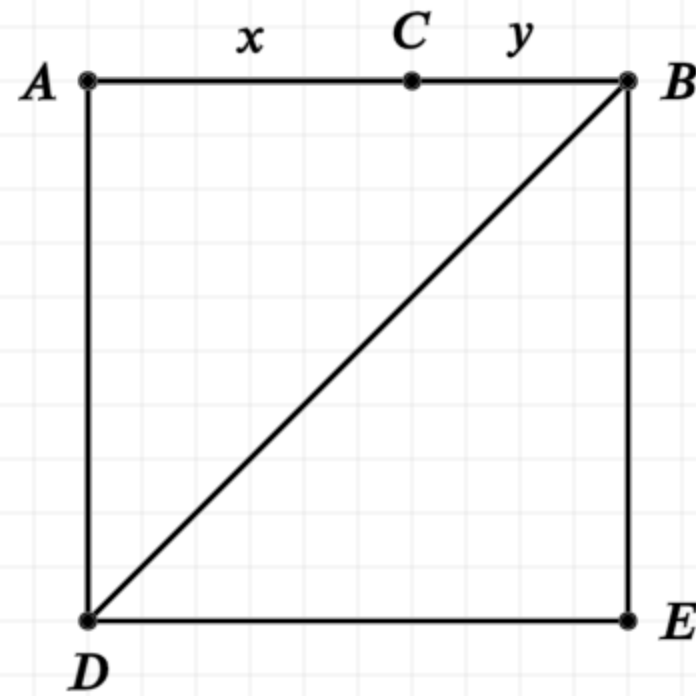
$$AB = AC + CB$$

$$AB^2 + BC^2 = AC^2 + 2 \cdot AB \cdot BC$$

$$(x + y)^2 + y^2 = x^2 + 2(x + y)y$$

Proposition 7 of Book 2

If a straight line be cut at random, the square on the whole and that of one of the segments both together are equal to twice the rectangle contained by the whole and the said segment and the square on the remaining segment.



In other words

Let AB be a straight line, arbitrarily cut at point C

$$AB^2 + BC^2 = AC^2 + 2 \cdot AB \cdot BC$$

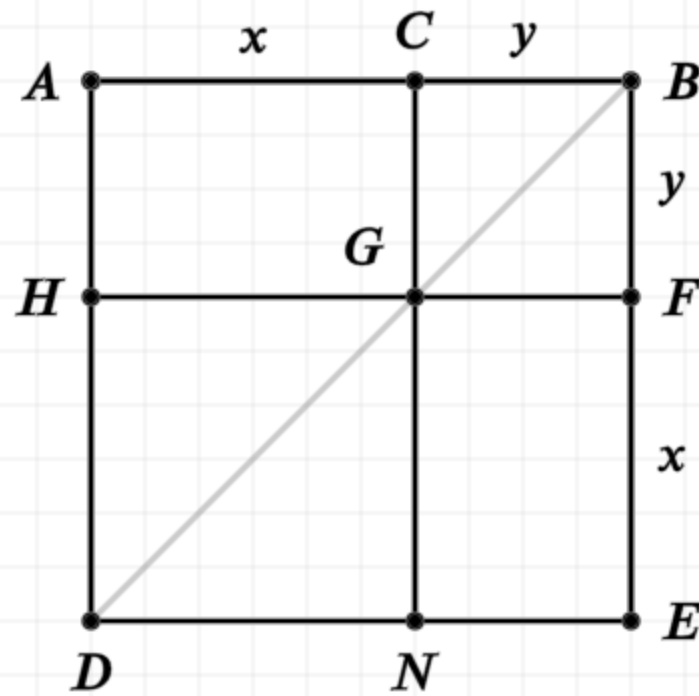
Construction:

Draw a square ADEB on the line AB (I.46), and draw the diagonal BD

$$AB = AC + CB$$

Proposition 7 of Book 2

If a straight line be cut at random, the square on the whole and that of one of the segments both together are equal to twice the rectangle contained by the whole and the said segment and the square on the remaining segment.



$$AB = AC + CB$$

In other words

Let AB be a straight line, arbitrarily cut at point C

$$AB^2 + BC^2 = AC^2 + 2 \cdot AB \cdot BC$$

Construction:

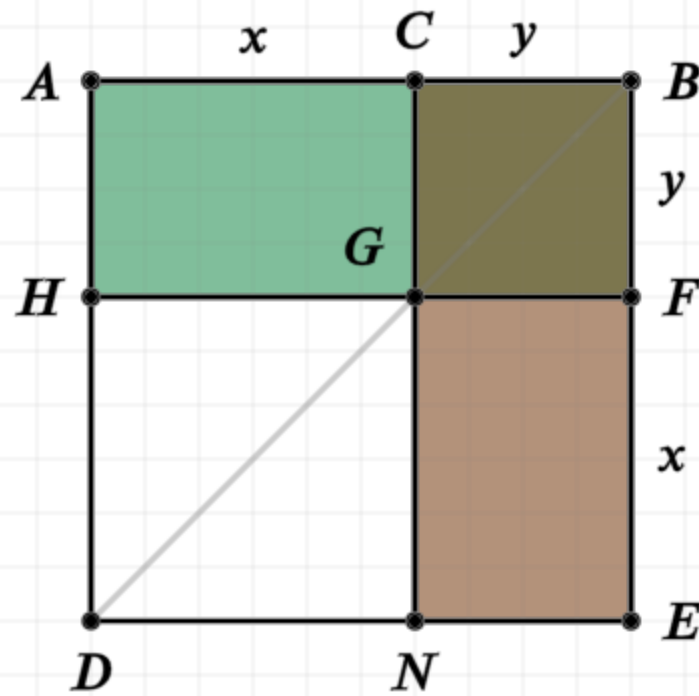
Draw a square ADEB on the line AB (I.46), and draw the diagonal BD

Draw a line CN parallel to AD (I.31), labelling the intersection with the diagonal as G

Draw a line parallel to AB through the point G (I.31).

Proposition 7 of Book 2

If a straight line be cut at random, the square on the whole and that of one of the segments both together are equal to twice the rectangle contained by the whole and the said segment and the square on the remaining segment.



In other words

Let AB be a straight line, arbitrarily cut at point C

$$AB^2 + BC^2 = AC^2 + 2 \cdot AB \cdot BC$$

Proof:

AG equals GE (I.43), add CF to both, thus preserving the equality

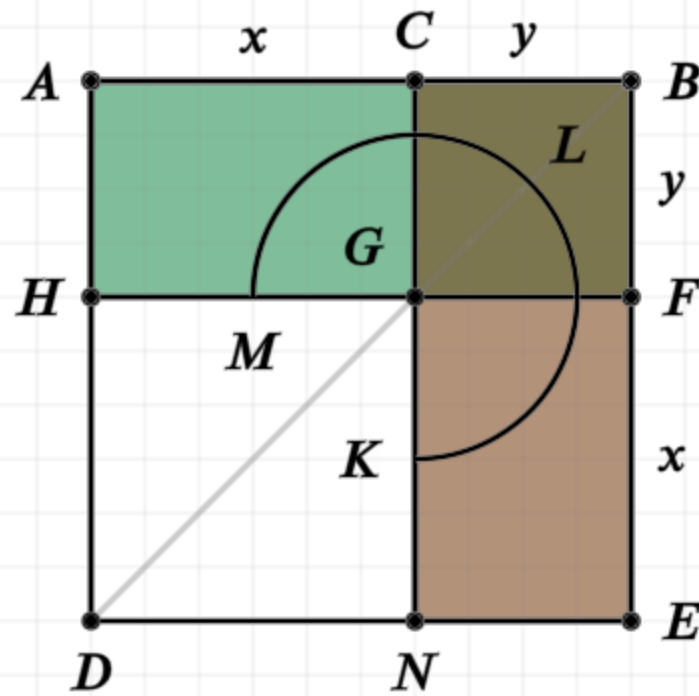
$$AB = AC + CB$$

$$\text{■} AF = \text{■} CE$$

$$\text{■} AF + \text{■} CE = 2 \text{■} AF$$

Proposition 7 of Book 2

If a straight line be cut at random, the square on the whole and that of one of the segments both together are equal to twice the rectangle contained by the whole and the said segment and the square on the remaining segment.



In other words

Let AB be a straight line, arbitrarily cut at point C

$$AB^2 + BC^2 = AC^2 + 2 \cdot AB \cdot BC$$

Proof:

AG equals GE (I.43), add CF to both, thus preserving the equality

But AF plus CE is equal to the gnomon KLM plus CE, therefore the gnomon KLM and CE is twice AF

$$AB = AC + CB$$

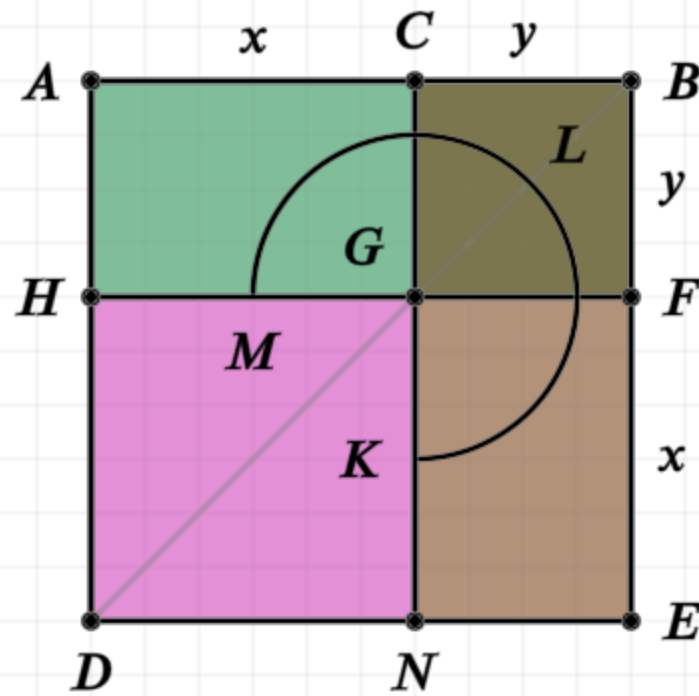
$$\text{AF} = \text{CE}$$

$$\text{AF} + \text{CE} = 2 \cdot \text{AF}$$

$$\text{KLM} + \text{CF} = \text{AF} + \text{CE} = 2 \cdot \text{AF}$$

Proposition 7 of Book 2

If a straight line be cut at random, the square on the whole and that of one of the segments both together are equal to twice the rectangle contained by the whole and the said segment and the square on the remaining segment.



In other words

Let AB be a straight line, arbitrarily cut at point C

$$AB^2 + BC^2 = AC^2 + 2 \cdot AB \cdot BC$$

Proof:

AG equals GE (I.43), add CF to both, thus preserving the equality

But AF plus CE is equal to the gnomon KLM plus CE, therefore the gnomon KLM and CE is twice AF

Let DG, the square on AC, be added to each

$$AB = AC + CB$$

$$\text{Green } AF = \text{Brown } CE$$

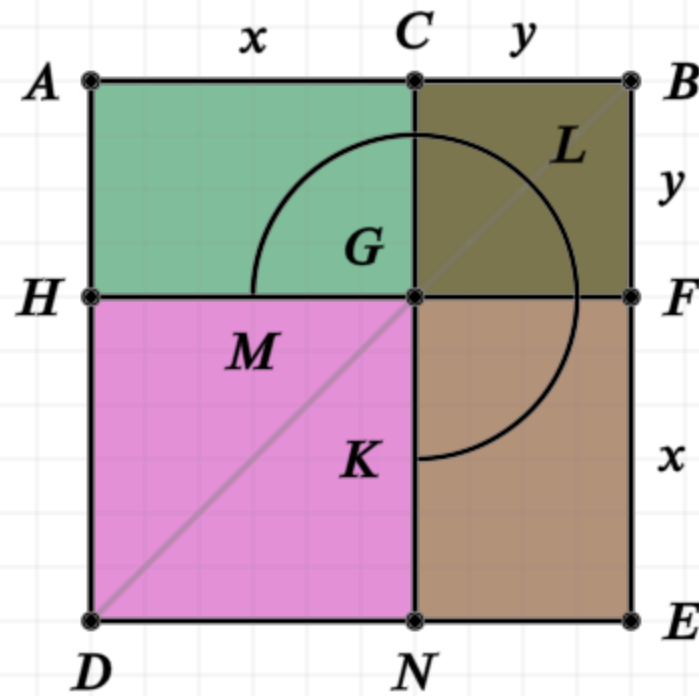
$$\text{Green } AF + \text{Brown } CE = 2 \text{ Green } AF$$

$$\text{Brown } KLM + \text{Brown } CF = \text{Green } AF + \text{Brown } CE = 2 \text{ Green } AF$$

$$\text{Brown } KLM + \text{Pink } DG + \text{Brown } CF = 2 \text{ Green } AF + \text{Pink } DG$$

Proposition 7 of Book 2

If a straight line be cut at random, the square on the whole and that of one of the segments both together are equal to twice the rectangle contained by the whole and the said segment and the square on the remaining segment.



In other words

Let AB be a straight line, arbitrarily cut at point C

$$AB^2 + BC^2 = AC^2 + 2 \cdot AB \cdot BC$$

Proof:

AG equals GE (I.43), add CF to both, thus preserving the equality

But AF plus CE is equal to the gnomon KLM plus CE, therefore the gnomon KLM and CE is twice AF

Let DG, the square on AC, be added to each

Let DG, the square on AC, be added to each

But the gnomon KLM and DG equals the square AE

$$AB = AC + CB$$

$$\text{green } AF = \text{brown } CE$$

$$\text{green } AF + \text{brown } CE = 2 \text{ green } AF$$

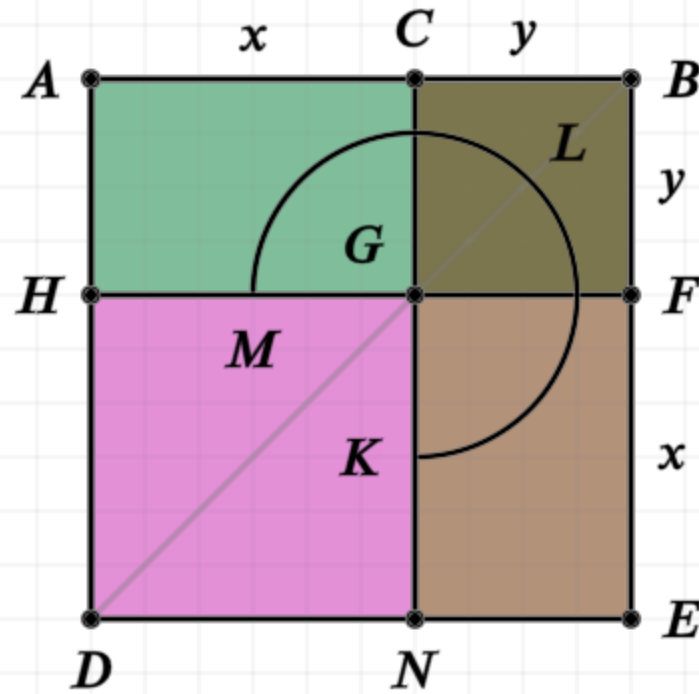
$$\text{brown } KLM + \text{brown } CF = \text{green } AF + \text{brown } CE = 2 \text{ green } AF$$

$$\text{brown } KLM + \text{pink } DG + \text{brown } CF = 2 \text{ green } AF + \text{pink } DG$$

$$\text{brown } AE + \text{brown } CF = 2 \text{ green } AF + \text{pink } DG$$

Proposition 7 of Book 2

If a straight line be cut at random, the square on the whole and that of one of the segments both together are equal to twice the rectangle contained by the whole and the said segment and the square on the remaining segment.



$$AB = AC + CB$$

$$\text{Area of } AF = \text{Area of } CE$$

$$\text{Area of } AF + \text{Area of } CE = 2 \cdot \text{Area of } AF$$

$$\text{Area of } KLM + \text{Area of } CF = \text{Area of } AF + \text{Area of } CE = 2 \cdot \text{Area of } AF$$

$$\text{Area of } KLM + \text{Area of } DG + \text{Area of } CF = 2 \cdot \text{Area of } AF + \text{Area of } DG$$

$$\text{Area of } AE + \text{Area of } CF = 2 \cdot \text{Area of } AF + \text{Area of } DG$$

$$AB^2 + CB^2 = 2 \cdot AB \cdot BC + AC^2$$

In other words

Let AB be a straight line, arbitrarily cut at point C

$$AB^2 + BC^2 = AC^2 + 2 \cdot AB \cdot BC$$

Proof:

AG equals GE (I.43), add CF to both, thus preserving the equality

But AF plus CE is equal to the gnomon KLM plus CE, therefore the gnomon KLM and CE is twice AF

Let DG, the square on AC, be added to each

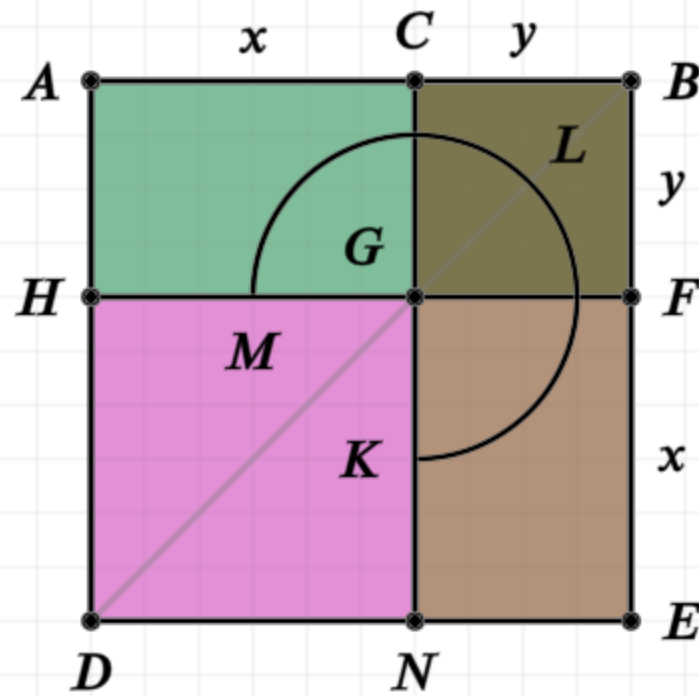
Let DG, the square on AC, be added to each

But the gnomon KLM and DG equals the square AE

AE equals the square on AB, CF equals the square on CB, AF is the rectangle formed by AB and BC, and finally DG is the square on AC

Proposition 7 of Book 2

If a straight line be cut at random, the square on the whole and that of one of the segments both together are equal to twice the rectangle contained by the whole and the said segment and the square on the remaining segment.



$$AB = AC + CB$$

$$\text{■ } AF = \text{■ } CE$$

$$\text{■ } AF + \text{■ } CE = 2 \text{■ } AF$$

$$\text{■ } KLM + \text{■ } CF = \text{■ } AF + \text{■ } CE = 2 \text{■ } AF$$

$$\text{■ } KLM + \text{■ } DG + \text{■ } CF = 2 \text{■ } AF + \text{■ } DG$$

$$\text{■ } AE + \text{■ } CF = 2 \text{■ } AF + \text{■ } DG$$

$$AB^2 + CB^2 = 2 \cdot AB \cdot BC + AC^2$$

In other words

Let AB be a straight line, arbitrarily cut at point C

$$AB^2 + BC^2 = AC^2 + 2 \cdot AB \cdot BC$$

Proof:

AG equals GE (I.43), add CF to both, thus preserving the equality

But AF plus CE is equal to the gnomon KLM plus CE, therefore the gnomon KLM and CE is twice AF

Let DG, the square on AC, be added to each

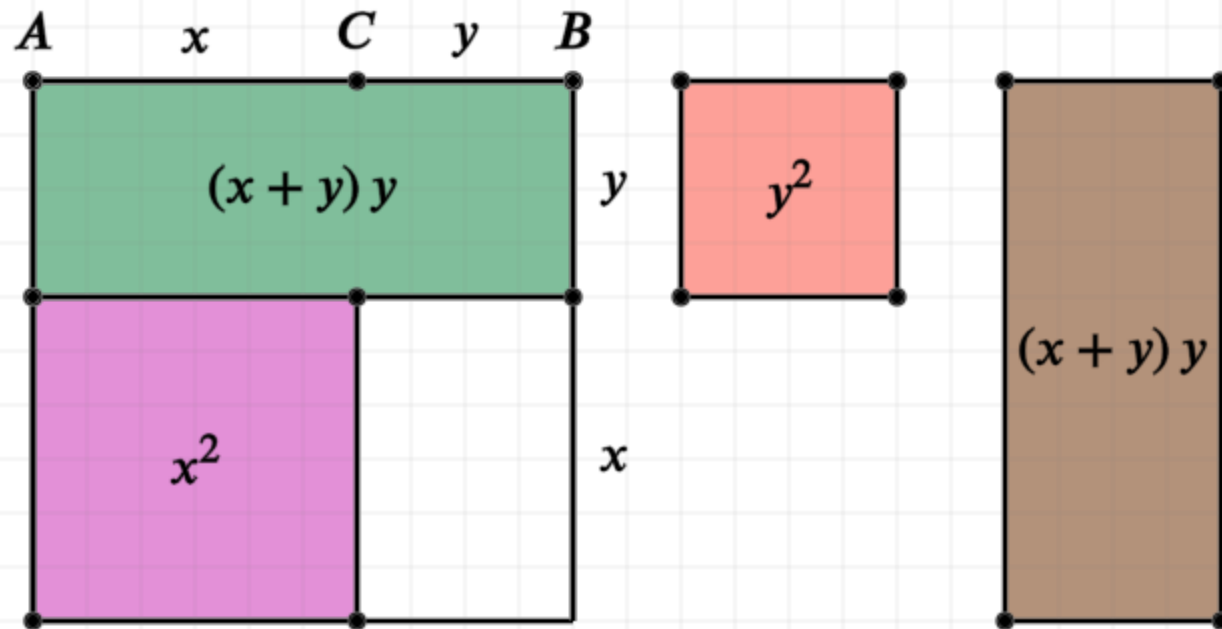
Let DG, the square on AC, be added to each

But the gnomon KLM and DG equals the square AE

AE equals the square on AB, CF equals the square on CB, AF is the rectangle formed by AB and BC, and finally DG is the square on AC

Proposition 7 of Book 2

If a straight line be cut at random, the square on the whole and that of one of the segments both together are equal to twice the rectangle contained by the whole and the said segment and the square on the remaining segment.



In other words

Let AB be a straight line, arbitrarily cut at point C

$$AB^2 + BC^2 = AC^2 + 2 \cdot AB \cdot BC$$

Proof:

AG equals GE (I.43), add CF to both, thus preserving the equality

But AF plus CE is equal to the gnomon KLM plus CE, therefore the gnomon KLM and CE is twice AF

Let DG, the square on AC, be added to each

Let DG, the square on AC, be added to each

But the gnomon KLM and DG equals the square AE

AE equals the square on AB, CF equals the square on CB, AF is the rectangle formed by AB and BC, and finally DG is the square on AC

$$AB^2 + BC^2 = AC^2 + 2 \cdot AB \cdot BC$$

$$(x + y)^2 + y^2 = x^2 + 2(x + y)y$$

Copyright © 2026 by Sandy Bultena

CC BY-NC-SA 4.0

The video and PDF content is licensed under Creative Commons Attribution-NonCommercial 4.0 International.
To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>

Content:

Youtube Videos: <https://www.youtube.com/c/SandyBultena>

PDFs: https://github.com/sandy-bultena/Euclid/tree/python-manim/Propositions_PDFs

Additional Credits

Source code to create videos:

© Alex Emily Oxorn - <https://github.com/sandy-bultena/Euclid/tree/python-manim>

This code could not have been created without the use of the manimgl libraries:

© 3Blue1Brown - <https://github.com/3b1b/manim>